

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and validated **greenhouse gas emission models** for diverse agricultural systems, translating complex agronomic science into structured logic for platform integration, achieving **95% methodology adherence** to IPCC standards.
- Collaborated with software developers to implement and test **GHG modelling logic** within the VSAg platform, ensuring accurate translation of scientific principles into functional code and contributing to a **15% reduction in model integration time**.
- Researched and incorporated national GHG inventory frameworks and emerging carbon accounting standards into agricultural models, ensuring compliance with evolving **regulatory frameworks** across multiple geographies.
- Automated data processing and reporting for agricultural datasets using **Python scripts**, streamlining the preparation of inputs for emission models and reducing manual effort by **30%**.
- Managed change control processes for updating system configurations and coordinating User Acceptance Testing (UAT) for interconnected enterprise business systems, ensuring smooth post-go-live support.
- Designed and implemented **Power BI dashboards** to visualize operational KPIs for agricultural datasets, translating complex data into actionable insights for stakeholders and improving report generation efficiency by **40%**.

SKILLS

Modelling & Analytics – Greenhouse Gas (GHG) Modelling, Carbon Intelligence, Quantitative Modelling, Statistical Analysis, Data Analysis, Agronomic Science, Agri-Environmental Science, Climate Action, IPCC Methodologies, National GHG Inventory Frameworks, Carbon Accounting Standards, A/B Testing, Time-Series Analysis, Machine Learning Fundamentals

Programming & Scripting – Python, R, Java, Spring Boot, SQL, Pandas, NumPy, Scikit-learn, Flask, FastAPI

Data Management & BI – Power BI, DAX, Power Query, Tableau, Data Warehousing, ETL, Database Modelling, PostgreSQL, MySQL, Oracle

Software Development & Operations – REST APIs, Microservices, Docker, AWS (EC2, S3, Lambda, CloudWatch), Linux, Shell Scripting, Git, Jenkins, CI/CD, Splunk, Dynatrace, CloudWatch

Agile & Project Management – Agile/Scrum, Jira, Confluence, Change Management, UAT Coordination, Incident Response, Production Support, SLA Management

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Agricultural GHG Emission Modelling Suite

- Developed a modular **GHG emission modelling suite** in Python, incorporating modules for tillage, dairy, and beef farming, adhering to **IPCC guidelines** and national inventory standards.
- Integrated **Pandas and NumPy** for efficient data manipulation and analysis of agricultural input data, processing datasets exceeding **10GB** for model simulations.
- Created interactive **Power BI dashboards** to visualize model outputs, presenting key emission factors and reduction scenarios to simulated stakeholders, enabling clearer comprehension of climate impact.

VSAg Platform Data Integration Pipeline

- Designed and implemented a **data pipeline** using Python and SQL to ingest and transform diverse agricultural datasets (e.g., soil types, fertilizer application, livestock data) for the VSAg platform.
- Ensured data quality and consistency through automated validation checks and error handling, reducing data discrepancy issues by **20%**.
- Collaborated with the development team to define API specifications for seamless integration of processed data into the core **VSAg software product**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)